

Report

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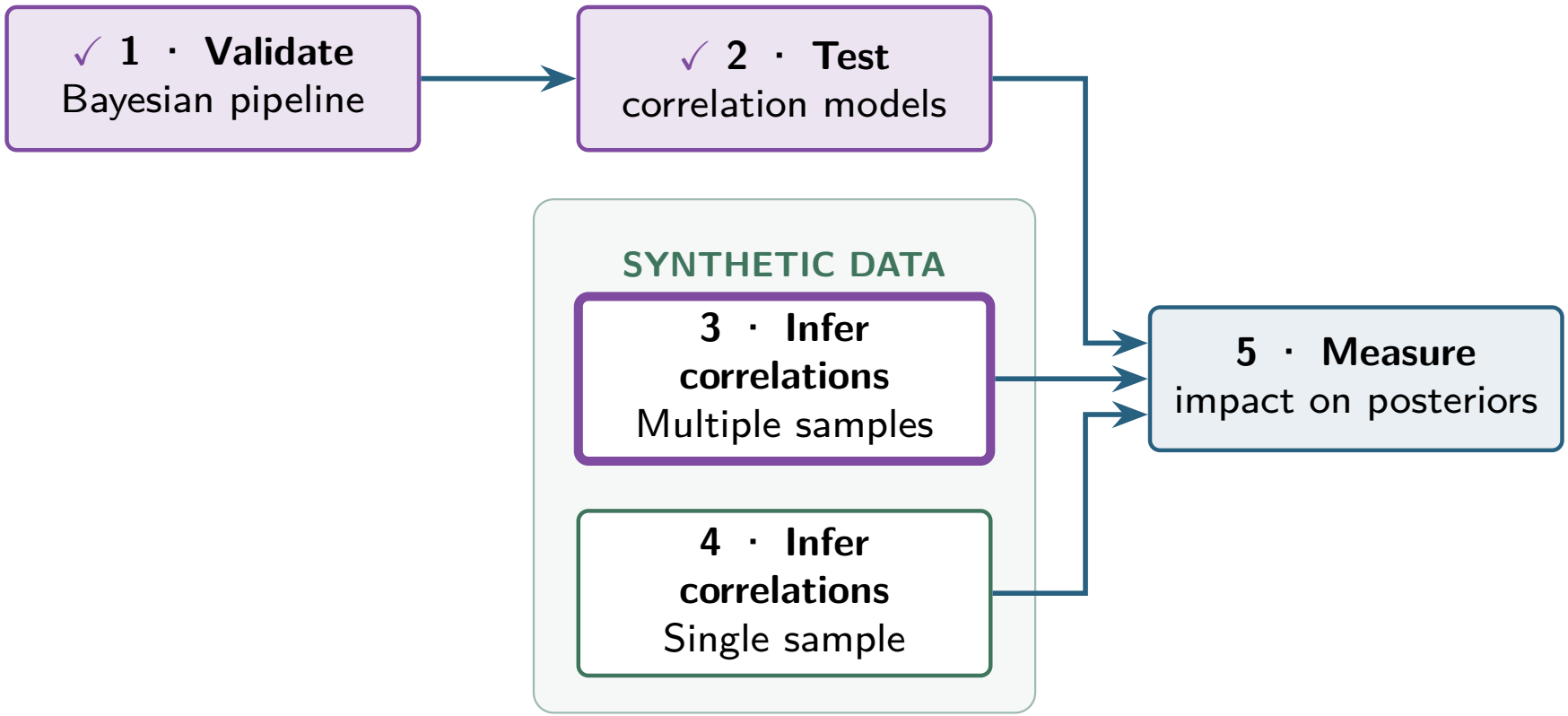
Summary

1 Tasks Status

Conference Materials Status

Task	Status	
2026-10 Conference	◔	Simpósio IFUSP Presentation
2026-10 Conference	◔	AGORÁ Presentation
2026-11 Poster	◔	3rd Schooll QC Poster
2026-12 Conference	○	LASNPA Presentation
2026-12 Paper	○	LASNPA Proceedings
2027-03 Conference	○	Quark Matter Presentation

Study Phases – Dependencies



Phase 1 – Bayesian Pipeline Validation

Task	**Status**	**Notebook**
↪ Reproduce Mankolli et al. results with the original JETSCAPE emulator	✓	Notebook
↪ Reproduce the same results with the `hic_bayes` emulator	✓	Notebook
↪ Emulator closure test	✓	Notebook
↪ MCMC convergence test	✓	Notebook

Phase 2 – Correlation Model Sensitivity

Task	Status	Output
↪ Au–Au-200 + d–Au-200: Correlation-model sensitivity		Posteriors
• Global CS correlation	✓	Notebook
• Global RBF correlation	✓	Notebook
• Global RBF correlation fix unit	✓	Notebook
• Sector CS correlation (central vs. combined laterals)	✓	Notebook
• Sector RBF correlation (central vs. combined laterals)	✓	Notebook
↪ Compare posterior shifts, credible intervals, and parameter correlations		Analysis
• Global correlation models	✓	Notebook
• Detector-grouped correlation models	◑	Notebook
↪ Quantify and compare the emulator-error and experimental-error covariance matrices in	◔	Analysis

Phase 3 – Correlation Inference from Multiple Samples

Task	**Status**	**Output**
↪ Formulate questions about the study	○	Questions
↪ Generate controlled synthetic datasets with known covariance (N samples; goal: $N = 10$)	◔	Dataset
↪ Classical correlation inference	○	Benchmark
↪ Quantum-learning correlation inference	○	Benchmark
↪ Classical vs. quantum comparison against the known covariance	○	Comparison

Phase 4 – Single-Sample Correlation Inference

Task	**Status**	**Output**
↪ Study correlation approximations when only one experimental sample is available	◔	Heuristics
↪ Use reported uncertainty components to constrain plausible correlation structures	○	Models

Phase 5 – End-to-End Bayesian Impact

Task	**Status**	**Output**
↪ Propagate inferred covariance matrices through the Bayesian likelihood • **Single sample:** CS, RBF, Zero, GP • **Multiple samples:** Classical, Quantum	◔	Posteriors
↪ Quantify covariance-estimation error → posterior error	○	Final study